

The Sensitivity of Archaeobacteria to Antibiotics

R. HILPERT, J. WINTER, W. HAMMES¹, and O. KANDLER

Received January 15, 1981

Summary

Most of the species of the two largest groups of archaeobacteria – halobacteria (9 strains) and methanogens (10 strains) – were tested by the agar diffusion test for their sensitivity to 28 well-known antibiotics including typical inhibitors of membrane function and the synthesis of cell wall polymers, protein or RNA. Archaeobacteria were found to be insensitive to many antibiotics acting against eubacteria and eucaryotes, e.g. those inhibiting the synthesis or cross-linkage of the peptide subunit of murein or the synthesis of RNA. They are, however, sensitive towards inhibitors of the lipid cycle of the biosynthesis of cell wall polymers, to the protein inhibitor chloramphenicol, and to the feed additives lasalocid and monensin, which interfere with the membrane function. It remains to be shown, whether this insensitivity to many of the antibiotics is due to the impermeability of the cytoplasmic membrane or to the inactivation of the antibiotic by the cell rather than to the lack of a particular target for the antibiotic. It is pointed out that the very characteristic sensitivity pattern of archaeobacteria may serve as a guideline in the search for further biochemical and molecular differences between archaeobacteria, eubacteria, and eucaryotes, and that it may also be helpful in designing selective media for the enrichment and isolation of new types of archaeobacteria or in controlling mixed populations of eubacteria, archaeobacteria, and eucaryotes, e.g. in the rumen and in biogas plants.

Key words: Archaeobacteria – *Halobacterium* – *Halococcus* – Methanogens – Antibiotic sensitivity – Monensin – Penicillin – Bacitracin – Rifampicin

Introduction

Archaeobacteria differ from the classical bacteria and the eucaryotic organisms with respect to many basic biochemical characteristics, e.g. the sequence of the ribosomal 16S RNA, cell wall polymers, membrane lipids, DNA-dependent RNA polymerase, etc. (Fox et al., 1980; Kandler, 1981). The biosyntheses or functions of compounds and structures unique to archaeobacteria might also serve as targets

¹ Present address: Institut für Lebensmitteltechnologie, Universität Hohenheim, Garbenstr. 25, D-7000 Stuttgart 70.

for antibiotics, whereas typical targets for antibiotics present in eubacteria and eucaryotes, respectively, may be absent. Hence, it is to be expected that the sensitivity of archaeobacteria towards antibiotics is quite different from that of the members of the other two kingdoms, as already exemplified in at least two distinct cases. Firstly, the inhibitors of murein synthesis, cycloserine, penicillin and vancomycin do not inhibit methane bacteria, thus reflecting the absence of murein in these organisms (Hammes et al., 1979), and, secondly, rifampicin and streptolydigin, potent inhibitors of the DNA-dependent RNA polymerase of eubacteria, do not inhibit the corresponding enzyme of archaeobacteria (Zillig et al., 1980).

A comparative study of the sensitivity of a wider range of archaeobacteria towards antibiotics may help to characterize other biochemical differences between the 3 kingdoms of organisms, although it must be emphasized that insensitivity to a particular antibiotic does not necessarily indicate the absence of the respective target but may merely be due to the inability of the organism to transport, or the ability to inactivate the particular antibiotic.

This paper details the sensitivity of members of most species of the methanogenic and obligate halophilic archaeobacteria towards 28 different antibiotics directed against various targets.

Materials and Methods

Organisms

The microorganisms employed and their sources are summarized in Table 1.

Media and culture conditions

The halophilic organisms were grown aerobically at 37 °C on agar medium made up of 3 solutions (Reistadt, 1970): solution A contains in 100 ml of distilled water: 200 g NaCl, 10 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ and 2 mg Fe (III) - citrate. Solution B contains in 400 ml of distilled water: 10 g pepton, 25 g glycerol and 9 g agar-agar. Solution C contains 50 g dry skimmed milk in 500 ml of distilled water and was pasteurized at 100 °C for 10 min. Solutions A and B were mixed thoroughly, the pH was adjusted to 8.4 and the mixture was autoclaved for 20 min. After cooling to 55 °C, solution C was added and agar plates were poured immediately. For the inoculation of petri dishes containing antibiotics, colonies from agar-plates were suspended in 20% NaCl solution.

All methanogenic organisms, besides the two thermophilic species and *Methanococcus vannielii*, were cultured at 37 °C in a modified MS-medium (Hammes et al., 1979). *Methanobacterium thermoautotrophicum* was grown at 62 °C in medium 2 of Balch et al. (1979). *Mc. vannielii* was grown in formate-medium according to Hippe (pers. comm.). The following ingredients were added to 360 ml of distilled water: 0.2 g KH_2PO_4 , 0.16 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 0.16 g NaCl, 0.16 g NH_4Cl , 20 mg $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$, 2 mg $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, 0.4 g yeast extract, 0.8 g Trypticase, 4 g sodium formate, 2.6 g NaHCO_3 , 8 g agar-agar, 2.4 ml trace minerals (Wolin et al., 1963), 4 ml vitamins (Wolin et al., 1963) and 0.4 ml resazurin (0.1% solution). After boiling under nitrogen, 16 ml of reducing agent (Hammes et al., 1979) was added. The medium was dispensed, in 100 ml portions, into serum bottles using an anaerobic glove box (Coy Laboratory Products Inc.). The stoppered bottles were autoclaved and agar-plates were poured anaerobically (Balch et al., 1979).

Mc. vannielii was grown on formate under nitrogen at 37 °C, the other methanogens under an atmosphere of 1.5 $\text{KN/m}^2\text{H}_2\text{:CO}_2$ (80:20%). Eubacteria and yeasts were grown aerobically at 30 °C in a medium consisting of 1% Trypticase, 0.5% yeast extract, 0.5% NaCl, 1.5% agar-agar, pH 7.5. For inoculation colonies were suspended in 0.9% NaCl solution.

Table 1. Microorganisms and their sources

Organisms	Source
<i>Halobacterium halobium</i> DSM 670	DSM ¹
<i>Halobacterium saccharovororum</i> DSM 1137	DSM ¹
<i>Halobacterium salinarium</i> DSM 668	DSM ¹
<i>Halococcus morrhuae</i> CCM 537	CCM ²
<i>Halococcus morrhuae</i> CCM 859	CCM ²
<i>Halococcus morrhuae</i> CCM 2226	CCM ²
<i>Halococcus morrhuae</i> CCM 2526	CCM ²
<i>Halobacterium spec. r-4</i>	F.R. Valera ³
<i>Methanobrevibacter arboriphilus</i> AZ DSM 744	R.S. Wolfe ⁴
<i>Methanobrevibacter smithii</i> PS DSM 861	R.S. Wolfe ⁴
<i>Methanobacterium bryantii</i> DSM 863	R.S. Wolfe ⁴
<i>Methanobacterium bryantii</i> M.o.H. G DSM 862	R.S. Wolfe ⁴
<i>Methanobacterium formicicum</i> DSM 1535	R.S. Wolfe ⁴
<i>Methanobacterium thermoautotrophicum</i> Δ H DSM 1053	R.S. Wolfe ⁴
<i>Methanobacterium thermoautotrophicum</i> 1	U. Temper ⁵
<i>Methanococcus vannielii</i> DSM 1224	DSM ¹
<i>Methanospirillum hungatei</i> J F 1 DSM 864	R.S. Wolfe ⁴
<i>Methanosarcina barkeri</i> MS DSM 800	R.S. Wolfe ⁴
<i>Escherichia coli</i> ATCC 11775	ATCC ⁶
<i>Bacillus subtilis</i> W 23	
<i>Micrococcus luteus</i> DSM 20030	DSM ¹
<i>Saccharomyces cerevisiae</i>	Collection of the Botanical
<i>Hansenula spec.</i>	Institute University of Munich

¹ DSM = Deutsche Sammlung von Mikroorganismen, Göttingen.

² CCM = Czechoslovakian Culture Collection of Microorganisms, Brno.

³ F.R. Valera, Facultad de Ciencias Y, Colegio Universitario de Alicante, Alicante, Spain.

⁴ Dr. R.S. Wolfe, Dept. of Microbiology, University of Illinois, Urbana, IL 61801, USA.

⁵ U. Temper, University of Munich, Botanical Institute, Menzinger Straße 67, D-8000 München 19 (Isolate from sludge fermenting at 55 °C).

⁶ ATCC = American Type Culture Collection, Rockville, Maryland/USA.

Antibiotics

The antibiotics were obtained from the suppliers given in parentheses: α -amanitin, amphotericin B, cycloheximid, erythromycin, gentamycin, gramicidin D, gramicidin S, kanamycin, kasugamycin, nonactin, oleandomycin, valinomycin, vancomycin (Sigma Chemical Company), tetracycline, flavomycin (Hoechst AG), D-cycloserine, lasalocid (Hoffmann-La Roche), rifampicin (Ciba-Geigy), polymyxin, penicillin G, monensin, virginiamycin, cephalosporin C (E. Lilly GmbH), fosfomycin (Boehringer, Mannheim), bacitracin, chloramphenicol (R. Paesel KG), enduracidin (Prof. Zähler, Tübingen), nisin (Koch & Light), subtilin (ICN, Pharmaceuticals), gardimycin (Lepetit), nocardicin A (Fujisawa Pharmaceutical CO).

Demonstration of growth inhibition by antibiotics

The agar diffusion test was applied for demonstrating the inhibitory effects of antibiotics. Aqueous solutions of antibiotics were prepared, except for gramicidin D, nonactin, monensin and lasalocid, which were solubilized in EtOH (10 mg/100 μ l EtOH). EtOH-stock-solutions were diluted with dist. H₂O to 1 mg/ml. Filter stripes (Schleicher & Schüll No. 2668, of 5 $\times$ 80 mm) were soaked with 350 μ l of antibiotic solution (1 mg/ml) and

placed across agar plates. Plates for eubacteria and yeasts were stored in a refrigerator for 2 h to allow the antibiotics to diffuse before inoculation. The inoculation of the medium for methanogens was carried out anaerobically in a glove box and plates were incubated anaerobically in stainless steel jars under 1.5 KN/m² H₂:CO₂ (80:20%) (Balch et al., 1979). All organisms were inoculated with a sterile microsyringe in a straight stroke perpendicular to the antibiotic strips using ca. 50 µl of a cell suspension or growing liquid culture, respectively. Yeasts, eubacteria and halophiles were incubated aerobically at 30 °C or 37 °C, respectively. The effect of antibiotics on the growth of the organisms was determined by measuring the zone of inhibition after the following intervals: yeasts and eubacteria after 1 and 2 days, halophiles after 6 and 14 days, mesophilic methanogens after 4 and 6 days and thermophilic methanogens after 3 and 5 days. The data in Table 2 refer to the second measurement.

Results and Discussion

The results of the agar diffusion test are shown in Table 2. The eubacterial and eucaryotic control organisms exhibit the expected responses, i. e. the yeasts are only sensitive to a few antibiotics (cycloheximid, gramicidin S, polymyxin and amphotericin B) whereas the eubacteria are susceptible to all antibiotics employed except cycloheximid. However, certain species of the eubacteria are not inhibited by some of the antibiotics interfering with membrane function. The archaeobacteria are not only insensitive to cycloheximid but also to at least half of the other antibiotics tested.

Inhibitors of cell wall synthesis

The results obtained with 12 antibiotics interfering with cell wall synthesis in eubacteria confirm earlier studies on methanogens (Jones et al., 1977; Kandler and Hippe, 1977; Hammes et al., 1979) and halophiles (Mescher and Strominger, 1975). Antibiotics interfering with the synthesis of the UDP-MurNAc-pentapeptide precursor of murein or with the cross-linking reactions (fosfomycin, D-cycloserine, vancomycin, penicillin, cephalosporin and nocardicin A) generally do not inhibit archaeobacteria with the exception of *Mc. vanniellii*, which is sensitive towards D-cycloserine, although this organism does not contain murein (Jones et al., 1977; Kandler and König, 1978) or any other known D-alanine-containing structure. In order to verify the observed inhibitory effect of D-cycloserine, the agar diffusion test was repeated several times and also the minimum inhibitory concentration of D-cycloserine in liquid formate medium inoculated with 1 % of a growing culture was determined. Complete and permanent growth inhibition was observed at a concentration of 0.3 mg/ml. At 0.1 mg/ml the inhibition was 63 % after 60 h, but the same final density as in the control was reached after 90 h of incubation. The addition of 5 mg/ml D-alanine did not prevent the inhibitory effect of D-cycloserine, as it does in eubacteria (Neuhaus, 1967).

The susceptibility of *Mc. vanniellii* to D-cycloserine was also observed by Jones et al. (1977), although the effect was distinctly weaker. A concentration of 1 mg/ml was required to cause 60 % inhibition. The target for D-cycloserine in *Methanococcus* remains to be elucidated. It may be unique to *Methanococcus*, since none of the numerous other methanogens or halophiles tested was D-cycloserine-sensitive.

Antibiotics interfering with the lipid cycle of murein biosynthesis in eubacteria

inhibit the members of the genera *Methanobacterium* and *Methanobrevibacter* (Table 2) which contain cell wall sacculi of pseudomurein (Kandler and König, 1978; Kandler, 1978). This indicates that lipid-activated precursors may also be involved in the biosynthesis of pseudomurein as discussed previously (Hammes et al., 1979). However, *Methanococcus* and the halophiles, which do not contain pseudomurein are also inhibited by bacitracin, known to form complexes with isoprenoyl pyrophosphate, thus preventing its dephosphorylation to the respective monophosphate which is essential in the synthesis of cell wall precursors (Stone and Strominger, 1971) and probably also in that of the diether lipids (Kates et al., 1970). Therefore, Mescher and Strominger (1975), the first to discover the susceptibility of various members of the genus *Halobacterium* to bacitracin, assumed that it may interfere with the synthesis of glycoprotein which forms the rigid cell envelope of these organisms. However, neither *Halococcus* nor *Methanococcus* possess such a glycoprotein envelope, although they are sensitive towards bacitracin (Table 2). Since both genera contain isoprenoid diether lipids like *Halobacterium*, the hypothesis of Basinger and Oliver (1979) based on studies of the effect of bacitracin on the lipid synthesis in *Hb. cutirubrum* is more likely. They suggest bacitracin to act "through inhibition of dephosphorylation of the isoprenoyl pyrophosphate precursor of this organism's diether lipids".

Gardimycin inhibits the same archaeobacteria as does bacitracin with the exception of *Methanococcus*, but the halophiles are distinctly less susceptible than the pseudomurein-containing methanogens. Similar to gardimycin (Somma et al., 1977), also enduracidin (Matsubashi et al., 1978) and nisin (Reisinger et al., 1980) interfere with the further utilization of lipid-activated murein precursors. Both antibiotics inhibit the same methanogens, but do not inhibit the halophiles, the only exception being *Hb. salinarium* which is slightly inhibited by nisin. It is remarkable that *Methanosarcina barkeri* is also distinctly susceptible to enduracidin, whereas it is insensitive to all the other antibiotics directed against cell wall synthesis. Subtilin, chemically related to nisin (Gross et al., 1973), exhibits an antimicrobial spectrum which is very similar to that of nisin. However, the inhibition of the pseudomurein-containing methanogens is less, that of *Mc. vannielii* and of *Hb. salinarium* is more pronounced.

Flavomycin, which also interferes with the formation of the glycan strands in an as yet unknown way in eubacteria, is almost inactive towards archaeobacteria with the remarkable exception of only one member of the methanogens and one of the halophiles tested. The susceptibility of *Hb. halobium* towards flavomycin was also reported by Mescher and Strominger (1975). The reason for such an erratic action remains to be elucidated, however.

In conclusion, the antimicrobial spectrum of the antibiotics directed against reactions of the lipid cycle of murein synthesis in eubacteria indicates that these antibiotics inhibit archaeobacteria by different mechanisms. They may interfere with lipid-activated precursors of carbohydrate-containing polymers (pseudomurein, glycoproteins etc.) or with the biosynthesis of isoprenoid diether lipids characteristic of archaeobacteria. The latter may be the target in the case of the sensitive halophiles and *Mc. vannielii*, whereas pseudomurein synthesis may be the most sensitive process in the *Methanobacteriaceae*. The frequent insensitivity of archaeobacteria to several of these antibiotics may be due either to the impermeability of the respective organism towards the particular antibiotic or to its inactivation by the organism rather than to the lack of a suitable target.

Table 2. Effect of antibiotics on the growth of methanogenic and halophilic bacteria (zone of inhibition (mm))

Target	Antibiotic	Archaeobacteria										Eubacteria					Eucaryotes			
		Halophiles					Methanogens					Eubacteria								
Cell wall	<i>Hb. halobium</i>	—	—	—	—	—	<i>Hb. spec. 1-4</i>	—	—	—	—	—	—	—	—	—	—	—	—	—
	<i>Hb. sachtarovorum</i>	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	<i>Hb. salinarum</i>	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	<i>Hc. mortuae</i> CCM 537	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	<i>Hc. mortuae</i> CCM 859	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	<i>Hc. mortuae</i> CCM 2226	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	<i>Hc. mortuae</i> CCM 2526	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	<i>Mbr. salinarum</i> AZ	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	<i>Mbr. smithii</i>	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	<i>Mbr. bryantii</i>	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	<i>Mb. bryantii</i> M.O.H.G.	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	Cell wall	<i>Mb. formicum</i>	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>Mb. thermoautotrophicum</i> ΔH		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>Mb. thermoautotrophicum</i> 1		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>Ms. vanielii</i>		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>Msp. hungatei</i>		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>Ms. barkeri</i>		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>E. coli</i>		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>Staph. aureus</i>		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>Bac. subtilis</i>		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>Mic. luteus</i>		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>Sacch. cerevisiae</i>		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
<i>Hansenula spec.</i>		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
RNA-Polymerase	α-Amanitin	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	Rifampicin	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Membrane-active antibiotics

Among the membrane-active antibiotics tested, monensin, a Na⁺-ionophore and lasalocid, a K⁺-ionophore are the most universal inhibitors of both, eubacteria and archaeobacteria. They are in use as feed additives in order to improve feed efficiency (Chen and Wolin, 1979). Although the direct inhibition of the methanogens of the rumen flora is doubted on the basis of experiments with the complex flora in rumen fluid (Chen and Wolin, 1979; Slyter, 1979) and because of the effect of the antibiotics on the physiological activity of washed-cell suspensions (Nevel and Demeyer, 1977), the growth inhibition of pure cultures of the methanogens is very pronounced (Table 2). Even *Msp. hungatei* and *Ms. barkeri*, not attacked by most of the other antibiotics are severely inhibited. The high sensitivity of methanogens towards monensin and lasalocid was also demonstrated in liquid media (Hilpert, Winter and Kandler, paper in preparation). Even concentrations as low as 1 µg/ml of monensin caused a 60 % lower optical density after 25 h of growth. However, full growth was observed after 60 h. This recovery after prolonged incubation was also observed at higher concentrations (100 µg/ml) causing complete inhibition for several days. The mechanism of the recovery process remains to be elucidated. The addition of monensin to fermenting sewage sludge caused an almost immediate decrease of methane production, an increase of volatile organic acids and a decrease in the number of methanogenic bacteria by more than one order of magnitude, thus indicating a direct effect on methanogens at least in the complex sludge fermentation.

All methanogens tested are also susceptible to gramicidin S, a cyclic decapeptide, which is known to increase the permeability of membranes in gram-positive bacteria. It does, however, not interfere with the growth of halophiles and *E. coli*. A very selective inhibition of *Mb. thermoautotrophicum* is caused by the two membrane-destructing antibiotics gramicidin D and polymyxin which are both inactive towards halophiles. No inhibition of any of the methanogens and no, or only erratic, effects on halophiles are found with the fungistatic amphotericin B and the ionophores nonactin and valinomycin.

Inhibitors of DNA-dependent RNA polymerases

Rifampicin is known to inhibit specifically the DNA-dependent RNA polymerase of eubacteria, whereas α -amanitin blocks the corresponding enzyme of eucaryotes (Jacob et al., 1970). Neither of the two antibiotics inhibits the DNA-dependent RNA polymerases of the various archaeobacteria, however (Zillig et al., 1980). Consequently, no inhibition of archaeobacteria was observed with α -amanitin. There was, however, a slight but significant inhibition of at least some of the halophiles and methanogens by rifampicin. The effect of rifampicin, at least in the case of the halophiles, was even more pronounced when the test was carried out in a liquid medium (Pecher and Böck, 1981). This may be due to a detergent effect rather than to the inhibition of the polymerases, however (Zillig et al., 1980).

Inhibitors of protein biosynthesis

Cycloheximid, a specific inhibitor of the protein synthesis of eucaryotic 80 S ribosomes does not inhibit any of the archaeobacteria tested, whereas chloramphenicol, a powerful inhibitor of the protein synthesis of eubacterial 70 S ribosomes inhibits all the methanogens, but none of the halophiles employed. The distinct inhibition of the methanogens by chloramphenicol does not necessarily indicate the presence of the same target as in eubacteria but may be due to the

pronounced chloramphenicol sensitivity of the hydrogenase involved in methane formation as described by McKellar and Sprott (1979). The absence of this enzyme in halophiles would explain their insusceptibility. However, inhibition of two species of *Halobacterium* was observed by Pecher and Böck (1981) employing liquid media instead of the agar diffusion test. A similar discrepancy between these two studies is observed in the case of tetracycline. Only *Mb. thermoautotrophicum* and *Mc. vannielii* show significant inhibition in the agar diffusion test (Table 2), whereas an even higher sensitivity of *Ms. barkeri* and two species of *Halobacterium* than of *Mc. vannielii* was observed by Pecher and Böck (1981). The different results may be due to the different osmotic stress exerted by the liquid and the solidified medium, even at the same salt concentration, as indicated by the observation that a slight damage of the cells may be overcome more easily on an agar surface than in a liquid surrounding. The distinct differences observed in the case of *Ms. barkeri* might also be caused by the use of different strains in the two studies.

Among the macrolide antibiotics only virginiamycin inhibits almost all strains of archaeobacteria tested, whereas oleandomycin and erythromycin are ineffective with the exception of a slight inhibition of *Mc. vannielii* by erythromycin. Of the two aminoglycoside antibiotics tested gentamycin inhibits most members of the *Methanobacteriaceae* and two of the genus *Halobacterium*, whereas kanamycin was without effect against any member of the archaeobacteria. The susceptibility of some of the methanogens towards several aminoglycoside antibiotics was also observed by Pecher and Böck (1981) whereas the two species of *Halobacterium* investigated in their study were insensitive to antibiotics of this type.

The characteristic pattern of susceptibility towards antibiotics observed in archaeobacteria may serve as a guideline for the enrichment and isolation of archaeobacteria from complex populations of natural habitats (Godsy, 1980) and for the search for additional biochemical and molecular differences between archaeobacteria, eubacteria and eucaryotes. It may also be of technological relevance, e. g. in controlling the mixed populations in the rumen and in biogas plants.

Acknowledgement. This work was supported by the Bundesministerium für Forschung und Technologie under Project PTB 8151.

References

- Balch, W.E., Fox, G.E., Magrum, L.J., Woese, C.R., Wolfe, R.S.: Methanogens: Reevaluation of a Unique Biological Group. *Bact. Rev.* 43, 260–296 (1979)
- Basinger, G.W., Oliver, J.D.: Inhibition of *Halobacterium cutirubrum* Lipid Biosynthesis by Bacitracin. *J. gen. Microbiol.* 111, 423–427 (1979)
- Chen, M., Wolin, M.J.: Effect of Monensin and Lasalocid-Sodium on the Growth of Methanogenic and Rumen Saccharolytic Bacteria. *Appl. Environ. Microbiol.* 38, 72–77 (1979)
- Fox, G.E., Stackebrandt, E., Hespell, R.B., Gibson, J., Maniloff, J., Dyer, T., Wolfe, R.S., Balch, W., Tanner, R., Magrum, L., Zablen, L.B., Blakemore, R., Gupta, R., Bonen, L., Lewis, B.J., Chen, K.N., Woese, C.R.: The Phylogeny of Prokaryotes. *J. Molec. Evol.* 13, 95 (1979)
- Godsy, E.M.: Isolation of *Methanobacterium bryantii* from a deep aquifer by using a novel broth-antibiotic disc method. *Appl. Environ. Microbiol.* 39, 1074–1081 (1980)
- Gross, H., Kiltz, H.H., Nebelin, E.: Subtilin, VI. Die Struktur des Subtilins. *Hoppe-Seyler's Z. physiol. Chem.* 354, 810–812 (1973)

- Hammes, W.P., Winter, J., Kandler, O.: The Sensitivity of the Pseudomurein-Containing Genus *Methanobacterium* to Inhibitors of Murein Synthesis. *Arch. Microbiol.* 123, 275–279 (1979)
- Jacob, S.T., Sajdel, E.M., Munro, H.N.: Specific action of α -amanitin on mammalian RNA polymerase protein. *Nature* 225, 60–62 (1970)
- Jones, J.B., Bowers, B., Stadtman, T.C.: *Methanococcus vannielii*: Ultrastructure and sensitivity to detergents and antibiotics. *J. Bact.* 130, 1357–1363 (1977)
- Kandler, O.: Zellwandstrukturen bei Methanbakterien. Zur Evolution der Prokaryonten. *Naturwissenschaften* 66, 95–105 (1979)
- Kandler, O.: Archaeobakterien und Phylogenie der Organismen. Vortrag anlässlich der Tagung der Gesellschaft Deutscher Naturforscher und Ärzte, Hamburg, Oktober 1980. Erscheint in *Naturwissenschaften* (1981)
- Kandler, O., Hippe, H.: Lack of peptidoglycan in the cell walls of *Methanosarcina barkeri*. *Arch. Microbiol.* 113, 57–60 (1977)
- Kandler, O., König, H.: Chemical composition of the peptidoglycan-free cell walls of methanogenic bacteria. *Arch. Microbiol.* 118, 141–152 (1978)
- Kates, M., Wassef, M.I., Pugh, E.L.: Origin of the glycerol moieties in the glycerol diether lipids of *Halobacterium cutirubrum*. *Biochim. Biophys. Acta* 202, 206–208 (1970)
- Matsubashi, M., Ohara, J., Yoshiyama, Y.: Inhibition of bacterial cell-wall synthesis in vitro by duracidin, a new polypeptide antibiotic. In: *Progr. Antimicrob. Anticancer Chemother.*, Proc. Inter. Congr. Chemother., 6th 1969, Tokyo, Vol. 1, p. 226. Baltimore, University Park Press 1970
- McKellar, R.C., Sprott, G.D.: Solubilization and Properties of a Particulate Hydrogenase from *Methanobacterium* Strain G2R. *J. Bact.* 139, 231–238 (1979)
- Mescher, M.F., Strominger, J.L.: Bacitracin induces sphere formation in *Halobacterium* species which lack a wall peptidoglycan. *J. gen. Microbiol.* 89, 375–378 (1975)
- Neuhaus, F.: D-cycloserine, as it is observed with eubacteria. In: *Antibiotics: Mechanism of Action*, Vol. 1, pp. 40–83, D. Gottlieb and P.D. Shaw, eds., Heidelberg, Springer-Verlag 1967
- Nevel van, C.J., Demeyer, D.I.: Effect of Monensin on Rumen Metabolism in Vitro. *Appl. Environ. Microbiol.* 34, 251–257 (1977)
- Pecher, T., Böck, A.: In vivo Susceptibility of Halophilic and Methanogenic Organisms to Protein Synthesis Inhibitors. *FEMS Lett.* (1981) in press
- Reisinger, P., Seidel, H., Tschesche, H., Hammes, W.P.: The Effect of Nisin on Murein Synthesis. *Arch. Microbiol.* 127, 187–193 (1980)
- Reistadt, R.: On the Composition and Nature of the Bulk Protein of Extremely Halophilic Bacteria. *Arch. Microbiol.* 71, 353–360 (1970)
- Slyter, L.L.: Monensin and Dichloroacetamide Influences on Methane and Volatile Fatty Acid Production by Rumen Bacteria in vitro. *Appl. Environ. Microbiol.* 37, 283–288 (1979)
- Somma, S., Merati, W., Parenti, F.: Gardimycin, a new antibiotic inhibiting peptidoglycan synthesis. *Antimicrob. Agents Chemother.* 11, 396–401 (1977)
- Stone, K.J., Strominger, J.L.: Mechanisms of action of bacitracin: complexation with metal ion and C₅₅-isoprenyl pyrophosphate. *Proc. nat. Acad. Sci. (Wash.)* 68, 3223–3227 (1971)
- Wolin, E.A., Wolin, M.J., Wolfe, R.S.: Formation of methane by bacterial extracts. *J. Biol. Chem.* 238, 2882–2886 (1963)
- Zillig, W., Stetter, K.O., Schulz, W., Janekovic, D.: Comparative Studies of Structure and Function of DNA-Dependent RNA Polymerase from Eubacteria and Archaeobacteria. In: *Enzyme Regulation and Mechanism of Action*, Vol. 60 (P. Mildner and B. Ries, eds.). Oxford–New York, Pergamon Press 1980